EMMA E. MCGEE, PHD, MSC

677 Huntington Avenue
Boston, MA 02115
+1 (707) 331-0273
emcgee@hsph.harvard.edu

RESEARCH INTERESTS

Cancer, causal inference, molecular data, pharmacoepidemiologic and lifestyle interventions, personalized medicine, dynamic treatment strategies, g-methods, target trial emulation

EDUCATION

2024 **PhD (Population Health Sciences - Epidemiology),** Harvard University, Boston, MA, USA
Concentrations: Biostatistics, Cancer
Thesis: Estimating the causal effects of dynamic strategies using observational data: Applications to cancer survivors

2020 **MSc (Epidemiology),** Harvard T.H. Chan School of Public Health, Boston, MA, USA
Concentration: Cancer Epidemiology

2015 **BA (Social Research and Public Policy),** Summa Cum Laude, New York University Abu Dhabi, Abu Dhabi, United Arab Emirates (UAE)
Minor: Literature

POSTDOCTORAL TRAINING

2024 – Present **Postdoctoral Research Fellow,** CAUSALab and Zhu Family Center for Global Cancer Prevention, Harvard T.H. Chan School of Public Health, Boston, MA, USA

OTHER ACADEMIC APPOINTMENTS

2025 – Present **Visiting Researcher,** Institute for Molecular Medicine Finland (FIMM), University of Helsinki, Helsinki, Finland

2024 – Present **Eric and Wendy Schmidt Center Collaborator,** Broad Institute of MIT and Harvard, Cambridge, MA, USA

2023 **Visiting Instructor,** University of Iceland, Reykjavik, Iceland

HOSPITAL APPOINTMENTS

2026 – Present **Big Data Health Scientist**, Veteran's Health Administration, Boston, MA, USA

2025 – Present **Consultant,** University of Texas Southwestern Medical Center, Dallas, TX, USA

2018 – 2024 **Research Associate,** Channing Division of Network Medicine, Brigham and Women's Hospital, Boston, MA, USA

OTHER PROFESSIONAL APPOINTMENTS

2017 – 2018 **Cancer Research Training Award Fellow,** National Cancer Institute, National Institutes of Health (NIH), Rockville, MD, USA

2015 – 2017 **Research Assistant in Cancer Epidemiology,** Pontificia Universidad Católica de Chile, Santiago, Chile

CURRENT RESEARCH FUNDING

2026 – Present **Veteran's Health Administration and National Institutes of Health/National Cancer Institute** (Big Data Scientist Training Enhancement Program)
Role: Big Data Health Scientist
Title: Personalized selection and sequencing of antiandrogen therapies for men with advanced prostate cancer
Aims: The first goal of this project is to compare the effectiveness of two of the most common antiandrogen agents for men with advanced prostate cancer, enzalutamide and abiraterone. The second goal of this project is to identify the optimal sequencing of these agents for different types of men.
Costs: $188,000

PAST RESEARCH FUNDING

2024 – 2025 **National Institutes of Health/National Institute on Minority Health and Health Disparities** (ZIA MD000022)
Role: Principal Investigator (Contract)
Title: Tailored exercise strategies and mortality among breast cancer survivors
Aims: This contract was for a principal investigator role on a target trial emulation of personalized physical activity strategies and breast cancer outcomes using data from Kaiser Permanente Pathways, a study that combines electronic health records with questionnaire data.
Costs: $100,000

2022 – 2024 **Broad Institute of MIT and Harvard/Eric and Wendy Schmidt Center**
Role: Predoctoral Fellow
Title: Estimating the causal effects of dynamic strategies using observational data
Aims: This fellowship recognized graduate students doing exceptional work in causal inference, machine learning, and human biology. The funded research focused on estimating the effects of dynamic lifestyle, treatment, and surveillance strategies for cancer survivors using observational data.
Costs: $100,000

2019 **Harvard T.H. Chan School of Public Health** (Rose Traveling Fellowship)
Role: Graduate Fellow

Title: Body fat composition and risk of breast cancer in older Icelandic women
Aims: This research fellowship was awarded by the chairs of the Departments of Epidemiology and Biostatistics to top postdocs, PhD students, and master's students. The funded proposal aimed to help cancer researchers get better answers to better scientific questions, with an application to body fat composition and breast cancer risk among adults in Iceland.
Costs: $5,000

PROJECTS SUBMITTED FOR FUNDING

Pending — **National Institutes of Health/National Cancer Institute** (K99/R00)
Role: Principal Investigator
Title: Personalizing breast cancer screening using innovative causal inference and artificial intelligence methods
Aims: The goal of this study is to answer questions about personalized breast cancer screening strategies by integrating multimodal, real-world data with causal inference and artificial intelligence methods.
Total Costs Requested: $1,017,000

Pending — **Patient Centered Outcomes Research Institute**
Role: Consultant
Title: Immunosuppression and cancer outcomes among kidney transplant recipients
Aims: The goal of this project is to evaluate the causal effects of immunosuppression strategies on cancer progression and allograft outcomes among kidney transplant recipients who have a post-transplant cancer diagnosis.
Total Costs Requested: $TBD

Costs are rounded to the nearest $1,000 US dollar.

HONORS AND DISTINCTIONS

SCIENTIFIC AWARDS

2024 — **Women in Medicine and Science Symposium Award,** Mass General Brigham

2024 — **Zhu Trainee Travel Award,** Harvard T.H. Chan School of Public Health ($2,000)

2023 — **Brian McMahon Travel Award,** Harvard T.H. Chan School of Public Health ($1,000)

2022 — **Brian McMahon Travel Award,** Harvard T.H. Chan School of Public Health ($1,000)

2020 — **Best Thesis Presentation,** Harvard T.H. Chan School of Public Health

2018 — **Best Oral Presentation,** Dana-Farber/Harvard Cancer Center Celebration of Early Career Investigators in Cancer Research

2017 — **Best Student/Fellow Poster**, National Cancer Institute Cohort Consortium Annual Meeting

2015	**Capstone Research Grant Award**, New York University Abu Dhabi ($1,000)
2014	**Summer Research Grant Award**, New York University Abu Dhabi ($5,000)

ACADEMIC AWARDS

2020	**Presidential Scholarship,** Harvard University ($18,000)
2020	**Dimitrios Trichopoulos Memorial Fund Award,** Harvard T.H. Chan School of Public Health ($2,000)
2018	**Loan Relief Award**, Harvard T.H. Chan School of Public Health ($27,000)
2015	**Summa Cum Laude,** New York University Abu Dhabi
2015	**University Honors Scholar/Founders' Day Award**, New York University
2014	**Rhodes Scholar Finalist,** US District 15, Rhodes Trust, University of Oxford
2011	**Full Scholarship**, New York University Abu Dhabi ($250,000)

TEACHING AWARDS

2022	**Bok Center Certificate of Distinction in Teaching,** Harvard University
2021 – 2022	**Excellence in Teaching,** Harvard T.H. Chan School of Public Health
2021	**Bok Center Certificate of Distinction in Teaching,** Harvard University
2020 – 2021	**Certificate of Distinction in Teaching,** Harvard T.H. Chan School of Public Health

BIBLIOGRAPHY

ARTICLES SUBMITTED OR UNDER REVIEW

1. **McGee EE**, Dickerman BA, Eliassen AH, Chen WY, García de Albéniz X, Hernán MA. Adjuvant bone-modifying agents and mortality among older women with early-stage breast cancer. *Submitted.*
2. **McGee EE**, García de Albéniz X, Eliassen AH, Yim K, Dickerman BA, Preston MA, Hernán MA. Cystoscopy surveillance and mortality in non-muscle invasive bladder cancer: A target trial emulation using real-world data. *Submitted.*
3. **McGee EE,*** Chalouni M,* van Santen DK, Stensrud MJ, Lodi S, Hernán MA. Causal survival analysis in the presence of unmeasured confounding for the effect of treatment on a competing event. *Under co-author review.*
4. **McGee EE,*** Amstutz A,* Madley-Dowd P, Hulme W, Gerlovin H, Robb W, Walker V, Palmer T, Katsoulis M, Takhar R, Denholm R, Sofat R, Hernán MA, Sterne JAC, Dickerman BA. Data assembly for causal inference from healthcare databases. *Under co-author review.*

5. **McGee EE,*** Lopes E,* et al. Guideline-based Healthy Diet and Lifestyle Interventions for the Prevention of Crohn's Disease: A Target Trial Emulation. *Under co-author review.*
6. Boca A, Torfadottir JE, Bellavia A, Tamimi RM, Mucci LA, Aspelund T, Gudnadson V, Chen WY, Valdimarsdottir UA, Dickerman BA, Eliassen AH, **McGee EE**. Getting better answers to better questions in cancer epidemiology: an application to body composition and breast cancer risk. *Under co-author review.*
7. Kurgansky K, **McGee EE**, Tuia J, Gerlovin H, Robb W, Fillmore N, Moll M, Cho K, Gaziano M, Hernán MA, Dickerman BA. mRNA-based Covid-19 vaccines and survival in patients receiving immune therapy for advanced cancer: A target trial emulation using electronic health records. *Under co-author review.*
8. Aherns E, Capdevilla M, Tenge T, Reimers JL, Sahil SD, Student SP, Wachtendorf LJ, Shay D, Cui J, Pensier J, **McGee EE,** Messinger CJ, Schaefer MS, O'Gara B. Effect of the primary anesthetic strategy on postoperative respiratory failure: A target trial emulation using real-world clinical data. *Submitted.*

PUBLISHED ARTICLES

1. Shincovich CI,* Chewcharat A,* Adomi M,* **McGee EE**, et al. Plasma Exchange and Daratumumab for Myeloma Cast Nephropathy. *American Journal of Hematology.* In press.
2. Jayasekera J, Ergas IJ, Schneider J, Tian E, Wojcik KW, Roh JM, Vo JB, Kushi LH, Wilson OWA, **McGee EE**. Tailored exercise strategies and mortality among breast cancer survivors. *JAMA Network Open.* 2026 Apr 13;9(4):e265177
3. Chen N, **McGee EE**, Nethery RC, Mucci LA, Dickerman BA. Guideline-based physical activity and health-related quality of life among prostate cancer survivors: a target trial emulation in the Health Professionals Follow-up Study. *American Journal of Epidemiology.* 2026:195(4):966-974.
4. **McGee EE**, Hernán MA, Giovannucci E, Mucci LA, Chiu YH, Eliassen AH, Dickerman BA. Estimating the effects of lifestyle interventions on mortality among cancer survivors: A methodological framework. *Epidemiology.* 2025;36(5):705-718.
5. Wang T, Zeleznik OA, **McGee EE,** Brantley KD, Balasubramanian R, Rosner BA, Willett WC, Avila-Pacheco J, Clish CB, Eliassen AH. Prospective study of circulating metabolomic profiles and breast cancer incidence among predominantly premenopausal women. *British Journal of Cancer*. 2025. doi: 10.1038/s41416-025-03159-2.
6. Khalili H, **McGee EE**, Challa PK, Roelstraete B, Johnell K, Schneeweiss S, Wastesson JW, Ludvigsson JF. Medications and risk of microscopic colitis: A nationwide study of older adults in Sweden. *Annals of Internal Medicine.* 2025;178(8):1106-1115.
7. Guo F, **McGee EE**, Chiu YH, Giovannucci E, Mucci LA, Dickerman BA. Evaluating recommendation-based dietary and physical activity strategies for prostate cancer prevention: A target trial emulation in the Health Professionals Follow-up Study. *American Journal of Epidemiology.* 2025;194(2):449-459.
8. **McGee EE**, Zeleznik OA, Balasubramanian R, Hu J, Rosner BA, Wactawski-Wende J, Clish CB, Avilla-Pacheco J, Willett WC, Rexrode KM, Tamimi RM, Eliassen AH. Differences in metabolomic profiles between Black and White women: Analyses from two prospective cohorts. *European Journal of Epidemiology.* 2024;39(6):653-665.
9. Koshiol J, Van De Wyngard V, **McGee EE**, Cook P, Pfeiffer RM, Mardones N, Medina K, Olivo V, Petit K, Jackson SS, Paredes F, Sanchez R, Huidobro A, Villaseca M, Bellolio E, Losada H, Roa JC, Hildesheim A, Araya JC, Ferreccio C. The Chile Biliary Longitudinal Study (Chile BiLS): A gallstone cohort. *American Journal of Epidemiology.* 2021;190(2):196-206.
10. **McGee EE**, Kim CH, Wang M, Spiegelman D, Stover DG, Heng YJ, Collins LC, Baker GM, Farvid MS, Schedin P, Jindal S, Tamimi RM, Eliassen AH. Erythrocyte membrane fatty acids and breast cancer risk by tumor tissue expression of immuno-inflammatory markers and fatty acid synthase. *Breast Cancer Research.* 2020;22(1):78.

11. **McGee EE**, Koshiol J. Comments on "Should we consider gallbladder cancer a new smoking-related cancer? A comprehensive meta-analysis focused on dose-response relationships" [Letter]. *International Journal of Cancer*. 2020;147(2):593-594.
12. Kotsopoulos J, **McGee EE**, Lozano-Esparza S, Garber JE, Ligibel JA, Collins LC, Polyak K, Brown MA, Narod SA, Tamimi RM, Eliassen AH. Premenopausal plasma osteoprotegerin and breast cancer risk: A case-control analysis nested within the Nurses' Health Study II. *Cancer Epidemiology, Biomarkers & Prevention*. 2020;29(6):1264-1270.
13. Lee MH, Gao YT, Huang YH, **McGee EE**, Lam T, Wang B, Shen MC, Rashid A, Pfeiffer RM, Hsing AW, Koshiol J. A metallomic approach to assess associations of serum metal levels with gallstones and gallbladder cancer. *Hepatology*. 2020;71(3):917-928.
14. **McGee EE,** Jackson SS, Petrick JP, Van Dyke AL, Adami HO, Albanes D, Andreotti G, Beane-Freeman LE, Berrington de González A, Buring JE, Chan AT, Chen Y, Fraser GE, Freedman ND, Gao YT, Gapstur SM, Gaziano JM, Giles GG, Grant EJ, Grodstein F, Hartge P, Jenab M, Kitahara CM, Knutsen SF, Koh WP, Larsson SC, Lee IM, Liao LM, Luo J, Milne RL, Monroe KR, Neuhouser ML, O'Brien KM, Peters U, Poynter JN, Purdue MP, Robien K, Sandler DP, Sawada N, Schairer C, Sesso HD, Simon TG, Sinha R, Stolzenberg-Solomon R, Tsugane S, Wang R, Weiderpass E, Weinstein SJ, White E, Wolk A, Yuan JM, Zeleniuch-Jacquotte A, Zhang X, Zhu B, McGlynn KA, Campbell PT, Koshiol J. Smoking, alcohol, and biliary tract cancer risk: A pooling project of 26 prospective studies. *Journal of the National Cancer Institute.* 2019;111(12):1263-1278.
15. **McGee EE**, Kiblawi R, Playdon MC, Eliassen AH. Nutritional metabolomics in cancer epidemiology: Current trends, challenges, and future directions. *Current Nutrition Reports*. 2019;8(3):187-201.
16. Jackson SS, Van Dyke AL, Zhu B, Pfeiffer RM, Petrick JL, Adami HO, Albanes D, Andreotti G, Beane-Freeman LE, Berrington de González A, Buring JE, Chan AT, Chen Y, Fraser GE, Freedman ND, Gao YT, Gapstur SM, Gaziano JM, Giles GG, Grant EJ, Grodstein F, Hartge P, Jenab M, Kitahara CM, Knutsen SF, Koh WP, Larsson SC, Lee IM, Liao LM, Luo J, **McGee EE**, Milne RL, Monroe KR, Neuhouser ML, O'Brien KM, Peters U, Poynter JN, Purdue MP, Robien K, Sandler DP, Sawada N, Schairer C, Sesso HD, Simon TG, Sinha R, Stolzenberg-Solomon R, Tsugane S, Wang R, Weiderpass E, Weinstein SJ, White E, Wolk A, Yuan JM, Zeleniuch-Jacquotte A, Zhang X, McGlynn KA, Campbell PT, Koshiol J. Anthropometric risk factors for cancers of the biliary tract in the biliary tract cancers pooling project. *Cancer Research.* 2019;79(15):3973-3982.
17. Echeverría G, Dussaillant C, **McGee EE**, Mena C, Nitsche MP, Urquiaga I, Bitran M, Pedrals N, Rigotti A. Promoting and implementing the Mediterranean diet in the Southern Hemisphere: The Chilean experience. *European Journal of Clinical Nutrition*. 2019;72(Suppl 1):38-46.
18. **McGee EE,** Castro FA, Engels EA, Freedman ND, Pfeiffer RM, Nogueira L, Stolzenberg-Solomon R, McGlynn KA, Hemminki K, Koshiol J. Associations between autoimmune conditions and hepatobiliary cancer risk among elderly US adults. *International Journal of Cancer*. 2019;144(4):707-717.
19. Koshiol J, Bellolio E, Vivallo C, Cook P, Roa JC, **McGee EE**, Losada H, Van Dyke AL, Van De Wyngard V, Prado R, Villaseca M, Riquelme P, Acevedo J, Olivo V, Petit K, Hildesheim A, Medina K, Memis B, Adsay V, Ferreccio C, Araya JC. Distribution of dysplasia and cancer in the gallbladder: An analysis from a high cancer-risk population. *Human Pathology*. 2018;82:87-94.
20. Liu Z, Kemp TJ, Gao YT, Corbel A, **McGee EE**, Wang B, Shen MC, Rashid A, Hsing AW, Hildesheim A, Pfeiffer RM, Pinto LA, Koshiol J. Association of circulating inflammation

proteins and gallstone disease. *Journal of Gastroenterology and Hepatology*. 2018;33(11):1920-1924.

21. Liu Z, Kemp TJ, Gao YT, Corbel A, **McGee EE**, Roa JC, Wang B, Araya JC, Shen MC, Rashid A, Hsing AW, Hildesheim A, C Ferreccio, Pfeiffer RM, Pinto LA, Koshiol J. Circulating levels of inflammatory proteins and survival in patients with gallbladder cancer. *Scientific Reports*. 2018;8(1):5671.
22. **McGee EE***, Echeverría G*, Urquiaga I, Jiménez P, D'Acuña S, Villarroel L, Velasco N, Leighton F, Rigotti A. Inverse associations between a locally validated Mediterranean diet index, overweight/obesity, and metabolic syndrome in Chilean adults. *Nutrients.* 2017;9(8):862.

*Co-first author

BOOK CHAPTERS

1. **McGee EE**, Hernán MA, Russo R (2027 - expected). Estimating the effects of sustained treatment strategies. In: *Pharmacoepidemiology, 7th Edition*. Edited by Brian Strom, Stephen Kimmel, Tobias Gerhard, and Vincent LoRe: Wiley-Blackwell.
2. Echeverría G, Dussaillant C, **McGee E**, Urquiaga I, Velasco N, Rigotti A (2017). Mediterranean diet beyond the Mediterranean basin: Chronic disease prevention and treatment. In: *Mediterranean Identities – Environment, Society, Culture*. Edited by Fuerst-Bjelis B. Rijeka: InTech.

EDITORIAL ROLES

AD HOC PEER REVIEWER

American Journal of Epidemiology
British Journal of Cancer
Cancer Epidemiology, Biomarkers, & Prevention
Epidemiology
European Journal of Epidemiology
International Journal of Cancer
Journal of Clinical Oncology (JCO)
Journal of the American Medical Association (JAMA)
JAMA Internal Medicine
Journal of the National Cancer Institute (JNCI)

SOFTWARE

R PACKAGES

'gfoRmula'
Implements the parametric g-formula algorithm of Robins (1986)
https://github.com/CausalInference/gfoRmula

SAS MACROS

'GFORMULA'
Implements the parametric g-formula algorithm of Robins (1986)
https://github.com/CausalInference/GFORMULA-SAS

INVITED PRESENTATIONS

NATIONAL AND INTERNATIONAL

2026 **Causal inference in cancer research** *(Oral presentation – Scheduled)*
Integrative Cancer Population Sciences Symposium, Tokyo, Japan

2026 **Estimating causal effects of surveillance interventions for cancer survivors** *(Oral presentation – Scheduled)*
22nd Kolokotrones Symposium, Boston, MA, USA

2026 **Causal survival analysis in the presence of unmeasured confounding for a competing event** *(Oral presentation)*
International Society for Pharmacoepidemiology (ISPE) Annual Meeting, Milan, Italy

2026 **From real-world data to real-world decisions: Target trial emulation in cancer research** (*Oral presentation)*
Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, Virtual

2026 **Tailored physical activity strategies and mortality among breast cancer survivors: a target trial emulation** *(Oral presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Phoenix, AZ, USA

2026 **Asking better questions, getting better answers: Target trial emulation for precision medicine** (*Oral presentation)*
Vanderbilt University Medical Center, Virtual

2026 **From real-world data to real-world decisions: Target trial emulation of individualized treatment strategies** *(Oral presentation*)
AstraZeneza, Virtual

2025 **The clone-censor-weight approach: From basic principles to its transformation in diverse exposure and outcome scenarios** *(Oral symposium)*
International Society for Pharmacoepidemiology (ISPE) Annual Meeting, Washington D.C., USA

2025 **Getting better answers to better causal questions: Target trial emulation of dynamic treatment strategies** (*Oral presentation)*
Kaiser Permanente Division of Research, Virtual

2025 **Causal survival analysis in the presence of unmeasured confounding for a competing event** *(Oral presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Boston, MA, USA

2025 **Methodological challenges in descriptive and causal analyses of early-onset disease** *(Oral presentation)*
Society for Epidemiologic Research (SER) Mid-year Meeting, Virtual

2024 **Target trial emulation of dynamic surveillance strategies for cancer survivors: An application to non-muscle invasive bladder cancer** *(Oral presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Austin, TX, USA

2024 **Comparative effectiveness of cystoscopy surveillance strategies on mortality in non-muscle invasive bladder cancer: A target trial emulation using real-world data** *(Poster presentation)*
American Association for Cancer Research (AACR) Annual Meeting, San Diego, CA, USA

2023 **Estimating the long-term effects of lifestyle interventions for cancer survivors using target trial emulation and dynamic treatment strategies** *(Oral presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Portland, OR, USA

2022 **Disparities in metabolomic profiles between Black and White women** *(Poster presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Chicago, IL, USA

2018 **The Chile Biliary Longitudinal Study (Chile BiLS): A cohort profile** *(Poster presentation)*
Society for Epidemiologic Research (SER) Annual Meeting, Baltimore, MD, USA

2017 **Managing requests for large data projects in REDCap** *(Oral presentation)*
REDCapCon, New York, NY, USA

LOCAL

2025 **Getting better answers to better questions: Applications of target trial emulation to cancer epidemiology** *(Oral presentation)*
Ovarian Cancer Epidemiology and Analytics Network (OCEAN) Meeting, Virtual

2024 **Adjuvant bone-modifying agents and mortality among older postmenopausal women with early-stage breast cancer** *(Poster presentation)*
Women in Medicine and Science Symposium, Boston, MA, USA

2024 **Estimating causal effects from observational studies when treatment is unknown at baseline** *(Oral presentation)*
Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard, Boston, MA, USA

2024 **Causal inference from human data: How do we learn what works?** *(Oral presentation)*
Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard, Boston, MA, USA

2023 **Methodological challenges in estimating the long-term effects of lifestyle interventions for cancer survivors** *(Oral presentation)*
Dana-Farber/Harvard Cancer Center Celebration of Early Career Investigators in Cancer Research, Boston, MA, USA

2023	**Identifying effective interventions for cancer survivors using causal inference methods and high-dimensional data** *(Oral presentation)* Population Health Sciences Research Talk, Harvard T.H. Chan School of Public Health, Boston, MA, USA
2023	**Precision medicine, health equity, and causal mediation: An application to metabolomic profiles among Black and White women** *(Poster presentation)* Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard, Boston, MA, USA
2022	**Disparities in metabolomic profiles between Black and White women: Findings from two prospective cohorts** *(Oral presentation)* Metabolomics Working Group, Channing Division of Network Medicine, Brigham and Women's Hospital, Boston, MA, USA
2020	**Racial differences in metabolomic profiles** *(Oral presentation)* Metabolomics Working Group, Channing Division of Network Medicine, Brigham and Women's Hospital, Boston, MA, USA
2019	**Erythrocyte membrane fatty acids and breast cancer risk by tumor tissue expression of immuno-inflammatory markers and fatty acid synthase** *(Oral presentation)* Dana-Farber/Harvard Cancer Center Celebration of Early Career Investigators in Cancer Research, Boston, MA, USA
2019	**Plasma osteoprotegerin and breast cancer risk: A nested case-control study within the Nurses' Health Study II** *(Oral presentation)* Dana-Farber/Harvard Cancer Center Breast and Gynecologic Cancer Symposium, Boston, MA, USA
2019	**Nutritional metabolomics in cancer epidemiology** *(Oral presentation)* Breast Cancer Working Group, Channing Division of Network Medicine, Brigham and Women's Hospital, Boston, MA, USA
2018	**Cigarette smoking, alcohol consumption, and biliary tract cancer risk: A pooling project of 26 prospective studies** *(Oral presentation)* Dana-Farber/Harvard Cancer Center Celebration of Early Career Investigators in Cancer Research, Boston, MA, USA
2018	**Hepatobiliary cancer risk among elderly US adults with a history of autoimmune disease: A registry-based case-control study** *(Oral presentation)* National Cancer Institute, Infections and Immunoepidemiology Branch, Rockville, MD, USA
2017	**Highlights of findings from the Biliary Tract Cancers Pooling Project (BiTCaPP)** *(Poster presentation)* National Cancer Institute Cohort Consortium Annual Meeting, Rockville, MD, USA

2016 **The Chile Biliary Longitudinal Study (Chile BiLS): A cohort study protocol**
(Poster presentation)
Advanced Center for Chronic Diseases (ACCDiS) Cancer Spring Symposium, Santiago, Chile

TEACHING

TEACHING AT HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH

2027 (scheduled) **Data Assembly for Causal Inference from Healthcare Databases**
Role: Lead Instructor
Audience: Students from academia, industry, clinical medicine, and government
Description: Leading a new public facing course on data assembly procedures for causal inference analyses from complex, longitudinal, healthcare databases.

2026 – Present **Study Designs for Epidemiologists**
Role: Lead Instructor
Audience: 39 master's students
Description: Serving as lead instructor (with Dr. Sonia Hernández-Díaz) for a core methods course in the MPH in Epidemiology program. Teaching advanced study design concepts for randomized trials, observational cohorts, and case-control and self-controlled studies.

2025 – 2026 **Models for Causal Inference**
Role: Guest Lecturer
Audience: 75 PhD and master's students
Description: Taught two 90-minute lectures for a core methods course in the PhD and MSc programs in Epidemiology, covering inverse probability weighting and marginal structural models.

2022 – 2026 **Target Trial Emulation**
Role: Curriculum Fellow and Head Teaching Fellow
Audience: ~100 students from academia, industry, clinical medicine, and government
Description: Developed course materials and taught four 80-minute sessions for a public facing course on target trial emulation and causal inference methods.

2021 – 2022 **Quantitative Research Methods in Population Health Sciences**
Role: Teaching Fellow
Audience: 40 PhD students
Description: Developed content, wrote problem sets and exams, and led lab sessions. Taught maximum likelihood estimation, generalized linear models, survival analysis, g-methods, sensitivity analyses, instrumental variables, missing data, and other methods.

2021 **Core Principles of Biostatistics and Epidemiology**
Role: Head Teaching Fellow
Audience: 180 PhD and master's students

Description: Led a team of 16 teaching fellows in the instruction of students from 10 different academic programs. Taught epidemiologic methods, study design, and analyses.

2020 – 2021 **Models for Causal Inference**
Role: Teaching Fellow
Audience: 75 PhD and master's students
Description: Led weekly lab sessions on g-methods for causal inference, instrumental variable estimation, propensity scores, and other methods.

OTHER TEACHING

2027 (scheduled) **Let's Get Personal: Defining and Estimating Effects of Dynamic Treatment Strategies using Real-world Data,** American Causal Inference Conference (ACIC), Pittsburgh, PA, USA
Role: Lead Instructor
Audience: Statisticians, epidemiologists, and other methodologists
Description: Developing and leading a short course on dynamic treatment strategies for personalized medicine at ACIC.

2026 **Let's Get Personal: Defining and Estimating Effects of Dynamic Treatment Strategies using Real-world Data,** International Society for Pharmacoepidemiology (ISPE) Annual Meeting, Milan, Italy
Role: Lead Instructor
Audience: ~200 conference attendees
Description: Developed and led a short course on dynamic treatment strategies for personalized medicine at the ISPE annual meeting.

2026 **Let's Get Personal: Defining and Estimating Effects of Dynamic Treatment Strategies using Real-world Data,** Society for Epidemiologic Research (SER) Annual Meeting, Phoenix, AZ, USA
Role: Lead Instructor
Audience: ~25 conference attendees
Description: Developed and led a short course on dynamic treatment strategies for personalized medicine at the SER annual meeting.

2023 **Seminar in Causal Inference: Target Trial Emulation,** University of Iceland, Reykjavik, Iceland
Role: Visiting Instructor
Audience: 25 faculty and graduate students
Description: Developed and led a short course on target trial emulation and causal inference methods.

SUPERVISION AND MENTORING

Period	*Name*	*Position and role*	*Current position*
2026 – Present	**Kurt Herzog**	Mentee position: Visiting PhD student in Epidemiology, Harvard T.H. Chan School of Public Health (Home institution: Charité Universitatsmedizin Berlin). Role as mentor:	--

		Primary mentor for doctoral thesis project on causal inference approaches for comparing different durations of cancer treatments	
2026 – Present	**Andra Boca**	Mentee position: Master's student in Epidemiology, Harvard T.H. Chan School of Public Health. Role as mentor: Primary advisor for summer research practicum on clarifying research questions in cancer epidemiology	--
2025	**Tiffany Hsieh**	Mentee position: PhD student in Epidemiology, Brown University. Role as mentor: External advisor for PhD proposal on causal inference and machine learning methods for cancer research	PhD student in Epidemiology, Brown University
2024 – 2025	**Nanjie Li**	Mentee position: Master's student in Epidemiology, Harvard T.H. Chan School of Public Health. Role as mentor: Co-advisor with Barbra Dickerman (primary mentor) for master's thesis on cardiovascular risk prediction among prostate cancer survivors	PhD Student in Epidemiology, Leibniz Institute for Prevention Research and Epidemiology
2021	**Gabriella Forchion**	Mentee position: Undergraduate student, Howard University. Role as mentor: Mentor for summer research position on metabolomics and breast cancer risk, CURE Program, Dana-Farber/Harvard Cancer Center	MD Candidate at Ponce Health Sciences University

PROFESSIONAL SOCIETIES

2025 – Present **International Society for Pharmacoepidemiology (ISPE),** Member

2021 – Present **Society for Epidemiologic Research (SER),** Member

2019 – Present **American Association for Cancer Research (AACR)**, Associate Member

SERVICE AND LEADERSHIP

2025 – Present **Leader,** CAUSALab Clinics causal inference consulting service for Boston area researchers, Harvard T.H. Chan School of Public Health

2025 **Session chair,** Society for Epidemiologic Research (SER) Mid-year Meeting

2022 – 2023 **Abstract Reviewer,** Society for Epidemiologic Research (SER) Annual Meeting

2021 – 2023	**President, Epidemiologic Methods Journal Club,** Harvard T.H. Chan School of Public Health, Boston, MA, USA
2020 – 2021	**Contact Tracing and Vaccine Outreach Volunteer (Spanish and English),** COVID-19 Academic Public Health Volunteer Corps (APHVC) and GOTVax, Boston, MA, USA
2019 – 2020	**Leader, Master's Program Committee,** Harvard T.H. Chan School of Public Health, Boston, MA, USA
2018 – 2020	**Member, Epidemiology Student Advisory Committee,** Harvard T.H. Chan School of Public Health, Boston, MA, USA
2018 – 2020	**Member, Harvard Chan Student Club on CANcer: Learn, Interact, Translate (CAN-LIT),** Harvard T.H. Chan School of Public Health, Boston, MA, USA
2017 – 2018	**Member, NIH Academy on Health Disparities,** NIH, Bethesda, MD, USA
2017 – 2018	**Leader, Community Outreach and Service Committee,** Fellows Committee, National Cancer Institute, NIH, Rockville, MD, USA
2014 – 2015	**Chair,** Public Health Think Tank Conference, New York University Abu Dhabi, Abu Dhabi, UAE

TECHNICAL SKILLS

Statistical Software and Markup	R, SAS, Python, Stata Markdown, LATEX
Epidemiologic Methods, and Statistics	Target trial emulation of dynamic (i.e., "personalized") strategies G-methods for causal inference (e.g., g-formula, inverse probability weighting, marginal structural models) Graphical causal models (DAGs, SWIGs) Analysis of real-world healthcare data (electronic health records, claims) Analysis of longitudinal cohort and case-control studies Integration of multimodal data into epidemiologic research Sensitivity analyses
Teaching	Didactic instruction in causal inference, epidemiology, and biostatistics Development of course materials Statistical coding instruction in R, SAS, and Stata
Research Management	Research documentation and reproducibility (GitHub) Staff and student training and supervision Development of standard operating procedures and study protocols Questionnaire design, testing, and implementation IRB submissions Research budgeting